



Snapshots of Nature-Based Recreation Across Rural Landscapes: Insights from Geotagged Photographs in the Northeastern United States

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Abstract

Nature-based recreation is of increasing economic importance to rural communities transitioning away from traditional natural resource extraction. Rural areas are rich in cultural ecosystem services (CES) that function as essential public goods, providing benefits to livelihoods and fostering conservation of large landscapes. Geotagged photographs from social media platforms capture detailed information on public use of natural areas that can be useful for stakeholders interested in promoting resilient social–ecological systems, but applications of such “big data” are often limited in either spatiotemporal or thematic scope. We integrated multiple aspects of crowdsourced image data to better understand human–environment interactions, focusing on the Northern Forest, a culturally and ecologically significant region of the northeastern United States. Images for the region were mined from Flickr, a crowdsourcing image platform, from 2012 to 2017 and assigned themes via a neural network–cluster analysis pipeline while geographic drivers of nature-based visitor engagement were assessed with random forest models to predict use of CES across the Northern Forest. Daily and seasonal patterns in photography were broadly consistent throughout the region, whereas annual photography trends were more variable. Eleven core themes were identified in images, with 70% of photographs depicting activities related to CES. Visitor engagement with nature was greater near roads and shorelines, and at higher elevations, reflecting tensions between accessibility and aesthetics for recreation. Automated image classification tools can rapidly extract relevant information from crowdsourced photography for exploring human–environment interactions, but researchers should consider multiple spatial and temporal scales in CES assessments of large social–ecological systems.

Keywords Content analysis · Cultural ecosystem services · Flickr · Machine learning · Social media · Image classification

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Introduction

Nature-based tourism and recreation provide vital public benefits to local and regional economies (Sonter et al. 2016; White et al. 2016) as well as human livelihoods (Daniel et al. 2012). These services help facilitate pro-environmental behaviors (Larson et al. 2018) and can generate funding and public support for conservation (Zaradic et al. 2009), especially in sparsely populated rural areas where external investments from visitation are valuable for protecting open spaces otherwise valued for conventional-use industries (Gössling 1999; Wunder 2000). Quantifying and categorizing visitor engagement is necessary for effectively managing demands for recreation and promoting sustainable use of open spaces (Eagles 2014), but conventional forms of visitation data (e.g., reported park attendance) are often limited in spatial scope and missing

detail on what activities and aesthetic values attract visitors (Wilkins et al. 2020). This limitation is particularly an issue in areas such as municipal parks and conservation easement properties where detailed information on patterns of use are not available.

Social media activity has been demonstrated as an effective indicator of nature-based tourism and recreation (Wood et al. 2013; Sessions et al. 2016; Tenkanen et al. 2017; Fisher et al. 2018), and a valuable tool for filling spatiotemporal gaps in land-use patterns over large regions. Geotagged crowdsourced photographs from online social media platforms provide detailed information on public use of natural areas that can be used for various social and ecological applications (Ghermandi and Sinclair 2019), including (1) mapping cultural ecosystem services (Oteros-Rozas et al. 2018; Clemente et al. 2019; Lee et al. 2019; Vaz et al. 2020), (2) identifying visitor preferences for different aesthetic and/or environmental features (van Zanten et al. 2016; Hausmann et al. 2017; Hausmann et al. 2018; Fisher et al. 2019; Mancini et al. 2019), (3) tracking spatial hotspots and trends in visitor engagement of protected and unprotected areas (Sonter et al. 2016; Mancini et al. 2018; Schirpke et al. 2018), (4) monitoring biodiversity and species of conservation concern (Medhat et al. 2017), and (5) quantifying economic value of cultural ecosystem services in remote areas (Sinclair et al. 2018).

Relationships between social media activity and empirically-measured visitation, while generally strong and positive, can vary among different types of visitor sites (Levin et al. 2017) and across geopolitical landscapes with different recording practices and underlying socioeconomic compositions (Kuehn et al. 2020); therefore, landscape assessments of nature-based recreation and tourism from geotagged photographs should be evaluated at multiple geopolitical scales. Geotagged photographs contain spatially, temporally, and thematically explicit information about nature-based tourism and recreation at individual visitor scales. While many studies on visitor engagement have exploited space and time characteristics of crowdsourced images, image content analyses remain limited and predominantly based on manually-labeled photographs, presenting some methodological challenges for synthesizing content from large volumes of social media data (Koylu et al. 2019; Lee et al. 2019; Egarter et al. 2021). Integrating multiple forms of social media metadata can help reveal new spatiotemporal patterns in visitor engagement and mobility across natural landscapes. We applied this perspective to better understand human–environment interactions in the Northern Forest Region of the northeastern United States (Kuehn et al. 2020)—a rural region where outdoor recreation is an important driver of local and regional economies. Specifically, we used image classification and clustering tools to identify content of

crowdsourced photographs and sought to answer the following questions:

1. What are the predominant themes of crowdsourced photographs in large rural regions and how do visitors engage with cultural-ecosystem services across time and space?
2. How well do different landscape characteristics explain human engagement with nature in rural spaces and predict suitability for visitation in distinct geopolitical regions?
3. How relevant are rural geotagged photographs to parks, protected areas, and outdoor recreation, and how well do unsupervised image classification algorithms track observed image content?

Following other studies that have classified themes of Flickr images (Richards and Tunçer 2018; Clemente et al. 2019; Lee et al. 2019; Vaz et al. 2020), we expected images to be dominated by landscape and organism-focused photography, with additional clusters related to outdoor recreation, people, and human structures. Additionally, we expected photography to be greatest during the daytime and between late spring and late fall, roughly matching regional visitation records from early May to the terminus of peak foliage. We hypothesized that model performance would be greater at the state level than the whole region (Kuehn et al. 2020) and engagement would be influenced by a combination of geographic variables, especially distance to roads and urban areas, where visitation would display a negative, linear relationship reflecting accessibility constraints on visitors, and topography and permanent water bodies, where visitation would display more nuanced, nonlinear relationships reflecting inherent conflicts between the accessibility and attractiveness of high vista points versus flatter coastal areas and lakesides. Finally, we predicted that the content of geotagged photographs in protected, public areas would be highly relevant to nature-based recreation, whereas the content of photos in private areas would be more broad and less relevant to cultural ecosystem services.

Methods

Study Region

Our study is focused on the Northern Forest Region (NFR) in the northeastern United States—a region that covers portions of four states (i.e., Maine, New Hampshire, Vermont, and New York; see Fig. 1). The 30-million-acre region is comprised of mountainous terrain, valleys, forests, and waterbodies that are home to an estimated two million residents (Northern Forest Center 2021). Although timber

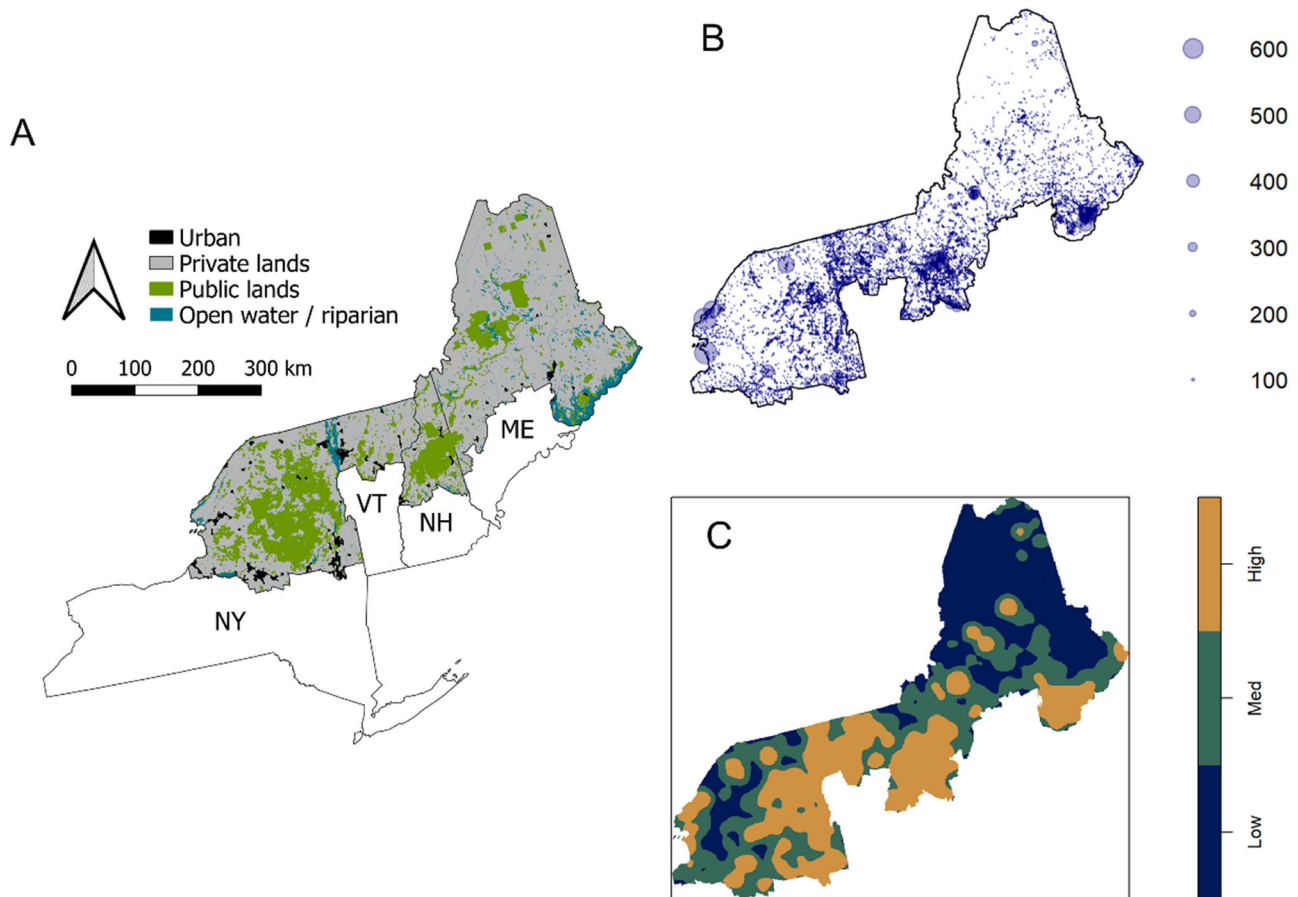


Fig. 1 Region of study (i.e., the Northern Forest Region) in the northeastern United States. **A** Landscape characteristics of the region, including urban areas (red), private areas (gray), public areas (green), and open waters and riparian corridors (blue). Urban areas were obtained from US Census TIGER files. Private and public zones were determined from the PAD-US database (Gergely and McKerrow 2013). Open waters and riparian corridors were obtained from the

National Hydrography Database, with open water constituting fresh-water ponds and lakes ≥ 10 acres, all estuarine and marine deep-water zones, and perennial rivers and streams. **B** Distribution of rural images, with point size proportional to image density per square kilometer. **C** Quantile-discretized kernel density surfaces of rural images, with densities categorized as low ($\leq 33\%$), medium (34–66%), and high ($\geq 67\%$) areas of Flickr activity

harvesting and wood product manufacturing are prevalent in the region, the tourism and recreation industry has been growing over the past decades and currently contributes an estimated \$14 billion each year into the region's economy (Northern Forest Center 2021).

Image Mining and Classification

Public images were obtained from Flickr, a web-based image repository and social media platform. Flickr data are effective for quantifying human visitation in nature in the NFR (Kuehn et al. 2020) and are routinely used by conservation researchers to understand human-environment interactions across diverse cultural and ecological spaces (Ghermandi and Sinclair 2019; Wilkins et al. 2020). Metadata for geotagged photographs in the NFR were obtained via Flickr's Application Programmer Interface (API). All publicly accessible images taken between 2012 and 2017 within the boundaries of the NFR were accessed.

Metadata for mined images included temporal and spatial information (date, time, latitude, and longitude for each photograph), publicly available photographer information (user ID, location of residence, personal image captions), as well as the associated ID and URL for each image. Image mining and metadata tabulation were performed with the Shapely (v1.5.17) and requests (v2.14.2) packages in the Python statistical software language (Van Rossum and Drake 2000) (all code available in supplemental GitHub repository at <https://github.com/hgoldspiel/flickr>).

Flickr images were characterized with "tags" using the Clarifai image annotation software (Zeiler and Fergus 2013; Clarifai 2016). Images were submitted to Clarifai's image labeling API to automatically characterize the content of the image. The Clarifai API uses computer vision techniques, including Deep Convolutional Neural Networks, to analyze images and return a list of objects and concepts ("tags") which it estimates are most probable to be present in each image. We used Clarifai's General model that recognizes a

Table 1 Geographic variables included in random forest models to predict visitor engagement in the Northern Forest Region

Geographic variable	Unit	Data Source
Elevation	meters	USGS National Elevation Database
Slope	percent (%)	USGS National Elevation Database
Roughness	index	USGS National Elevation Database
Distance to road	meters	US Census Bureau (TIGER files)
Distance to permanent water shores	meters	USGS National Hydrology Database
Distance to small urban areas (<100 K people)	meters	US Census Bureau; Statistics Canada
Distance to medium urban areas (100K–250K people)	meters	US Census Bureau; Statistics Canada
Distance to large urban areas (>250 K people)	meters	US Census Bureau; Statistics Canada
Land cover	type	USGS NLCD 2016

wide range of concepts and objects. Each image was submitted to the API in sequence and the top 10 tags were recorded.

Spatial Content Analysis

We performed a cluster analysis on the Clarifai-generated tags to identify principal themes of visitation in the region and distinguish among different forms of cultural-ecosystem services and other content conveyed by photographers. Specifically, we used a Monte Carlo network analysis to hierarchically cluster co-occurring image tags and determine an optimal number of clusters for assigning themes to photos, following methods described by Lee et al. (2019) (additional details in Appendix). The predominant theme of each image was determined according to the most common cluster of tags appearing in the list of ten tags assigned to images by Clarifai. Thematic “ties” were resolved by randomly assigning one of the competing themes to the image. Thematically similar clusters (e.g., food, beverages, dining) were grouped together to create a more concise breakdown of cultural and ecological themes in the Flickr dataset.

We used random forests (Breiman 2001) to evaluate how different geographic variables (Table 1) explain Flickr activity and predict human engagement with nature at multiple spatial scales. Random forests are a nonparametric machine-learning technique widely used to build predictive models and examine the importance of different covariates for social and ecological point patterns and processes, including for crowdsourced photography data (Richards and Tunçer 2018; Clemente et al. 2019; Egarter et al. 2021). For this analysis, we used distinct (non-duplicated), geolocated images and randomly selected one photograph per user per day (i.e., PUD), stratified by image theme, to reduce biases by individual users on spatiotemporal patterns of visitation (Wood et al. 2013; Egarter et al. 2021). We focused only on summer activity (May 1–September 30), as this season captures peak visitation in the study region and

was previously used to model empirically-measured park visitation records (Kuehn et al. 2020). We additionally focused our analysis on images representing cultural ecosystem services (i.e., those with assigned themes related to nature and landscape aesthetics). We treated image posts as presence points, randomly generated points ($n = 10,000$) as pseudo-absences, and linked presence/absence points to environmental variables for building models for each state and the entire NFR. The probability of visitor engagement with CES features (i.e., CES photograph occurrence) was modeled as a function of nine geographic covariates, including topographic features (elevation, slope, roughness), distance to linear features (roads, permanent water shorelines), distance to urban areas (small, medium, and large cities), and land cover (Table 2). Land cover data from the National Land Cover Database were reclassified to their eight major groups (1 = open water/ice/snow, 2 = developed, 3 = barren, 4 = forest, 5 = shrubland, 6 = herbaceous, 7 = planted/cultivated, and 8 = wetlands) prior to linking to photographs. All geographic layers were converted to rasters on the same spatial resolution (30 m) in QGIS (v3.16.1; (QGIS Development Team 2021) and linked to Flickr images in the R statistical software (R Core Team 2020) with the “extract” function from the *raster* package (Hijmans 2021). We assessed potential collinearity among covariates using variance inflation factors (VIFs; Zuur et al. 2009).

We fit our models in the *ranger* package in R (Wright and Ziegler 2017). For each model, we split our data into training (70%) and testing (30%) data and grew forests of 500 “trees” from the training data. We optimized models by tuning two hyperparameters, selecting a combination of values for the number of variables randomly selected when growing each tree (*mtry*) and the number of images randomly sampled from the training data for each tree (*sampsiz*) that minimized cross-validated error. Model sensitivity, specificity, and overall accuracy was calculated from a confusion matrix of observed and predicted values from the testing data in the *caret* package (Kuhn 2008);

Table 2 Themes of Clarifai-assigned image tags from rural images in the Northern Forest Region.

Theme	Images	Tags	Top ten tags
scenery ^a	88,635	606	sea, winter, lake, snow, river, loch, reflection, rock, shore, freezing
people	16,002	295	fun, girl, child, smile, event, people, male, sunglasses, sitting, glasses
biota ^a	11,548	125	flora, fauna, wildlife, mammal, bird, flower, beak, snout, spring, organism
structures	11,127	434	house, home, cottage, property, building, real estate, window, residential area, architecture, roof
transport	10,873	192	vehicle, car, road, motor vehicle, transport, infrastructure, mode of transport, lane, land vehicle, automotive exterior
aquatics ^a	3691	519	waterway, boat, water transportation, boating, boats and boating equipment and supplies, watercraft, kayak, dock, sailing, harbor
food/dining	2634	117	food, drink, cuisine, meal, lunch, eating, dish, animal source foods, supper, brunch
sports	1665	76	sports, team, team sport, games, competition, sport venue, player, ball game, football player, soccer
dogs	1620	270	dog like mammal, dog, dog breed group, dog breed, carnivoran, karelian bear dog, border collie, dog crossbreeds, sporting group, retriever
arts	756	64	performance, entertainment, performing arts, stage, musician, music, musical instrument, concert, audio, rock concert
equestrian	233	29	horse, horse like mammal, mane, mare, stallion, mustang horse, horse harness, bridle, horse supplies, rein

^aIndicates themes representing cultural ecosystem services

The top ten most important tags (ranked by eigenvectorcentrality) in each theme are listed

these cross-validation metrics were extended to test how models performed across different spatial scales. We used the Boruta algorithm (Kursa and Rudnicki 2010) to estimate and classify the importance of predictor variables to nature-based engagement. The algorithm implements an additional randomization procedure in random forest models, comparing the importance of variables from the original global model against those obtained from models with the variables randomly shuffled ($n = 500$ permutations) using a two-sided test ($\alpha = 0.01$). Partial dependence functions (i.e., marginal effects) were computed for each geographic variable in the *pdp* package (Greenwell 2017) to aid interpretation of relationships between covariates and engagement. Finally, we used our models and geographic layers to predict nature-based engagement and visitation suitability (i.e., models trained on images from public zones) in rural areas throughout the NFR and contrast insights obtained at different geopolitical scales.

Validation of Automated Image Tags

Given the central importance of crowdsourced data and automated image classification in this study, three techniques were used to validate the tag assignment and recreation relevance of Flickr images in the region. (1) The relevance of images to cultural ecosystem services was assessed with a sample of photographs from protected areas ($n = 198$). Images were independently reviewed by two researchers to determine whether the visual subject matter of images was related to nature-based recreation and the percent of agreement between researchers' responses was calculated. (2) To assess the overall relevance of top-ranking tags to images, a class of undergraduate students at

SUNY College of Environmental Science and Forestry was divided into two groups of 45 individuals and separately exposed to samples of 99 images and their top-ranked tag. Students were given 10 s per slide to independently determine whether each tag represented each image with a simple "yes/no" prompt and overall percent agreement for each image and tag was calculated for both groups. (3) We reviewed how well Clarifai tags matched the subjective text captions of photos with a random sample of captioned images ($n = 100$). Captions were qualitatively analyzed for content and compared to the Clarifai-generated tags using a two-phase inter-coding process (Cresswell 2009; Freelon 2010). In the first phase, all of the captions provided by photographers for all sample images were reviewed and "coded" according to the content (i.e., meaning) of the captions. The second phase involved identifying if the codes generated from the captions reflected the Clarifai-generated tags for the images. The percent agreement between content codes and Clarifai-tags was calculated for each tag and the overall sample.

Results

Image Mining and Classification

A total of 280,503 images were mined from Flickr in the Northern Forest Region between 2012 and 2017. Of those, 194,682 (69.4%) occurred in rural areas and 85,921 (30.6%) occurred in urban areas. More rural images occurred within private or inaccessible properties (118,179 or 60.7%) than in publicly accessible, protected areas (81,621 or 39.3%). Most photos were taken during the day (generally between

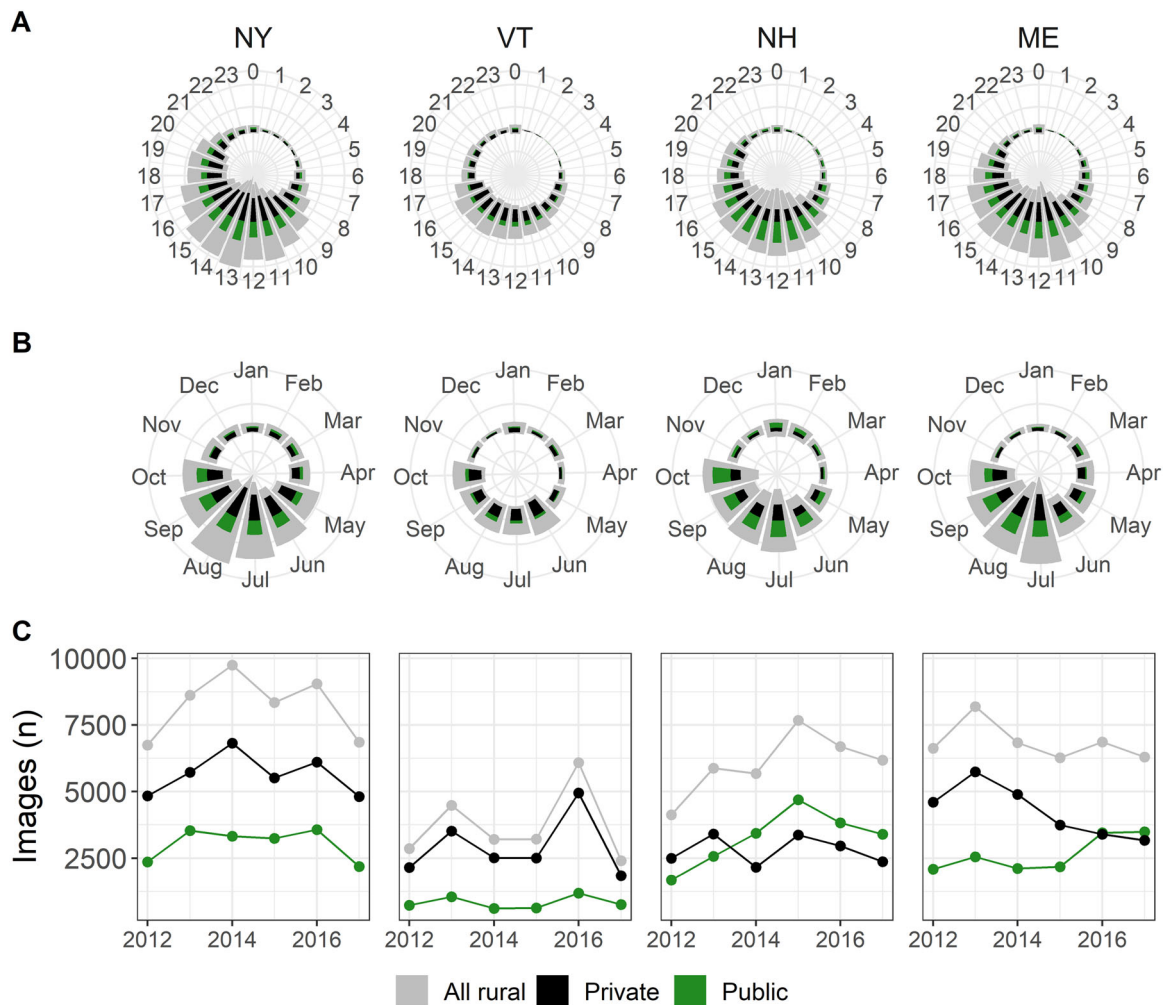


Fig. 2 Temporal trends in Flickr photography in the Northern Forest Region. Hourly (A), and monthly (B), and annual (C) variation in rural photography are shown for different states in the Northern Forest Region from 2012 to 2017. The number of images taken in public (green) and private (black) rural areas are shown, with the total for each place and time shown in gray

9 A.M. and 8 P.M.; Fig. 2A) from late spring to early fall (Fig. 2B), with the greatest image traffic in the summer from June to August when activity in public areas was particularly pronounced; these temporal and seasonal patterns were similar among states, with some regional differences (e.g., greatly reduced activity in Vermont after 5 PM; proportionally more peak activity in public areas than private areas in New Hampshire). Annual variation in volume of photographs was greater within states than the region as a whole, with some notable spikes in select years (e.g., 2016 in Vermont; Fig. 2C, Table S1). Some states (e.g., New Hampshire and Maine) displayed contrasting annual trends in private versus public areas, with images taken in public areas outranking those taken in private areas in select years. Total annual rural photography in the Northern Forest Region over the six years ranged from a minimum of 20,353 images in 2012 to a maximum of 28,684 images in 2016 (1.4-fold increase; Table S1). Aside from these annual fluctuations and state-level patterns (e.g., a positive trend in

photography in New Hampshire), there were no obvious trends over time at the regional scale. Clarifai assigned a total of 5219 unique tags to images in our rural dataset. Of these, 2081 tags (40%) were very rare, appearing in less than five images, and 14 tags (i.e., tree, sky, water, wilderness, nature, plant, mountain, cloud, vegetation, grass, nature reserve, mountainous landforms, body of water, and leaf) were very common, each appearing in over 10% of all fully tagged images. Many rural images (19%) were assigned fewer than 10 tags by Clarifai, with 240 photographs not receiving any tags at all.

Spatial Content Analysis

We identified multiple themes in rural images across the NFR, three of which were directly related to cultural ecosystem services (scenery, biota, and aquatic recreation). Modularity of image tag networks was optimized at 0.283 ($q_{0.025} = 0.242$, $q_{0.975} = 0.303$), with 7 steps and 26 clusters

Fig. 3 Content of rural images in the Northern Forest Region. Narrow black bars show the proportion of private (A) and public (B) images assigned to each aggregated tag cluster (i.e., theme). The overall content of rural images (private + public) is shown as gray bars in each figure

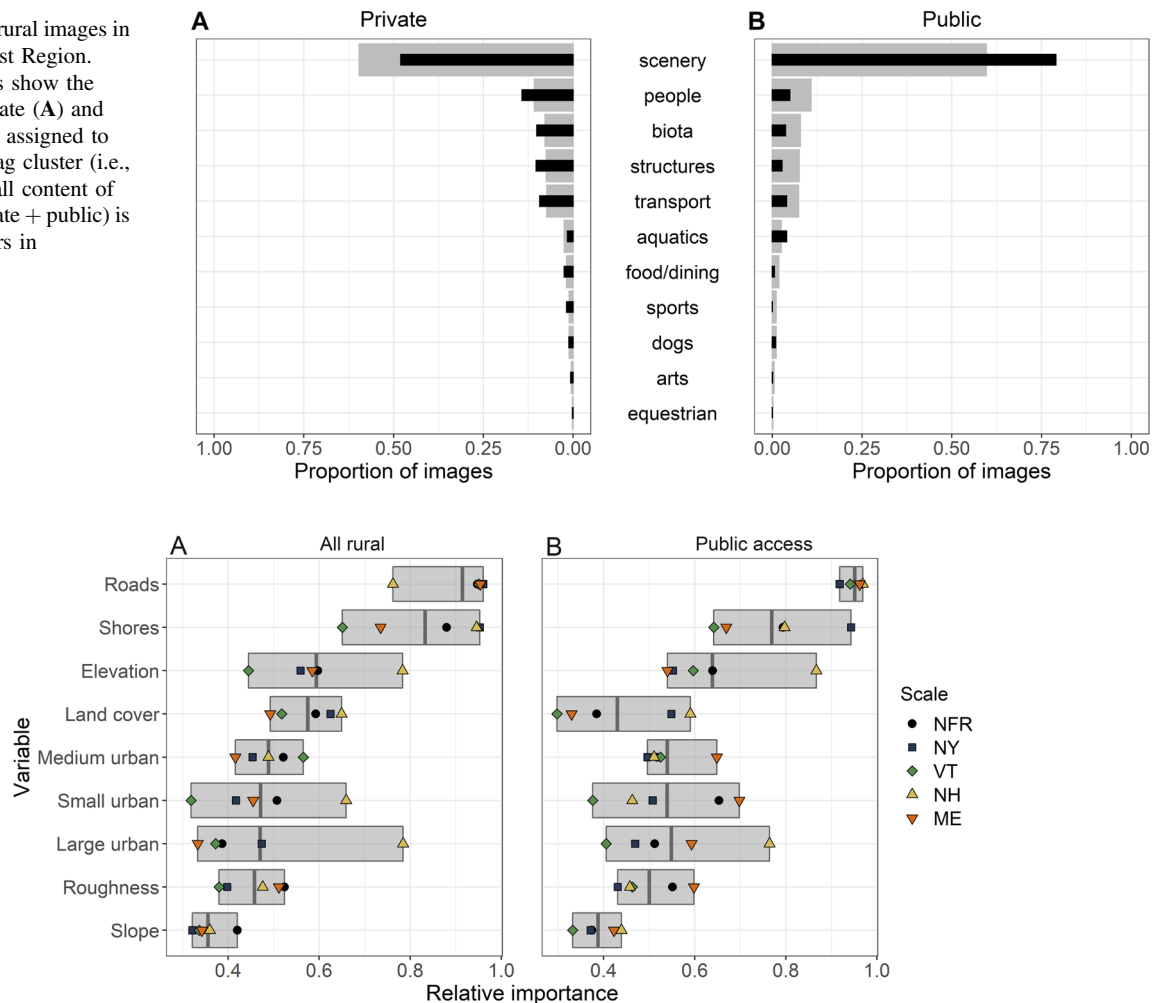


Fig. 4 Relative importance of geographic variables to summer CES images in the Northern Forest Region from 2012 to 2017. Importance rankings are shown for models fit with (A) all rural images or (B) only images in publicly accessible areas. Relative variable importance (mean importance of a variable divided by the max importance of any given variable) was calculated from random forest models run through

the Boruta algorithm at different geopolitical scales. The mean importance across all scales is shown for each variable as a gray vertical line. A description of variables can be found in Table 1 (Roads = distance to roads, Shores = distance to shorelines of permanent waterbodies)

(Fig. S2). Those 26 tag clusters were manually grouped according to content similarities, resulting in 11 core themes (Table 2). Most images (59.6%) were thematically categorized in a large cluster that primarily consisted of natural landscape content (i.e., “scenery”). Other key themes include people (10.7%), biota (7.7%), structures (7.5%), and transport (7.3%). Less common image themes included aquatic recreation (2.5%), food and dining (1.8%), sports (1.1%), dogs (1.1%), arts (0.5%), and equestrian activities (0.2%) (Fig. 3). Images assigned to the three themes pertaining to cultural ecosystem services constituted roughly 70% of the entire dataset. There were some differences between the types of content captured by people in private versus public areas (Fig. 3) and in different states (Fig. S3). Landscape scenery was more prominently featured by

people in public areas. Photos of people, biota, human structures, and transportation were more common in private areas (Fig. 3).

A total of 14,683 rural images were retained for spatial analyses on summer CES photography. All independent variables were verified as “important” drivers of nature-based engagement by the Boruta algorithm, but the relative importance of different features varied across geopolitical scales (Fig. 4). Distance to roads and shores had the greatest influence on human-environment interactions, with people less likely to engage with CES features at farther distances from roads (>1000 m) and shorelines (>50 m) (Figs. 5G, H, 6G, H). Topography was less important to engagement, but elevation displayed an important signature overall and for certain states in public

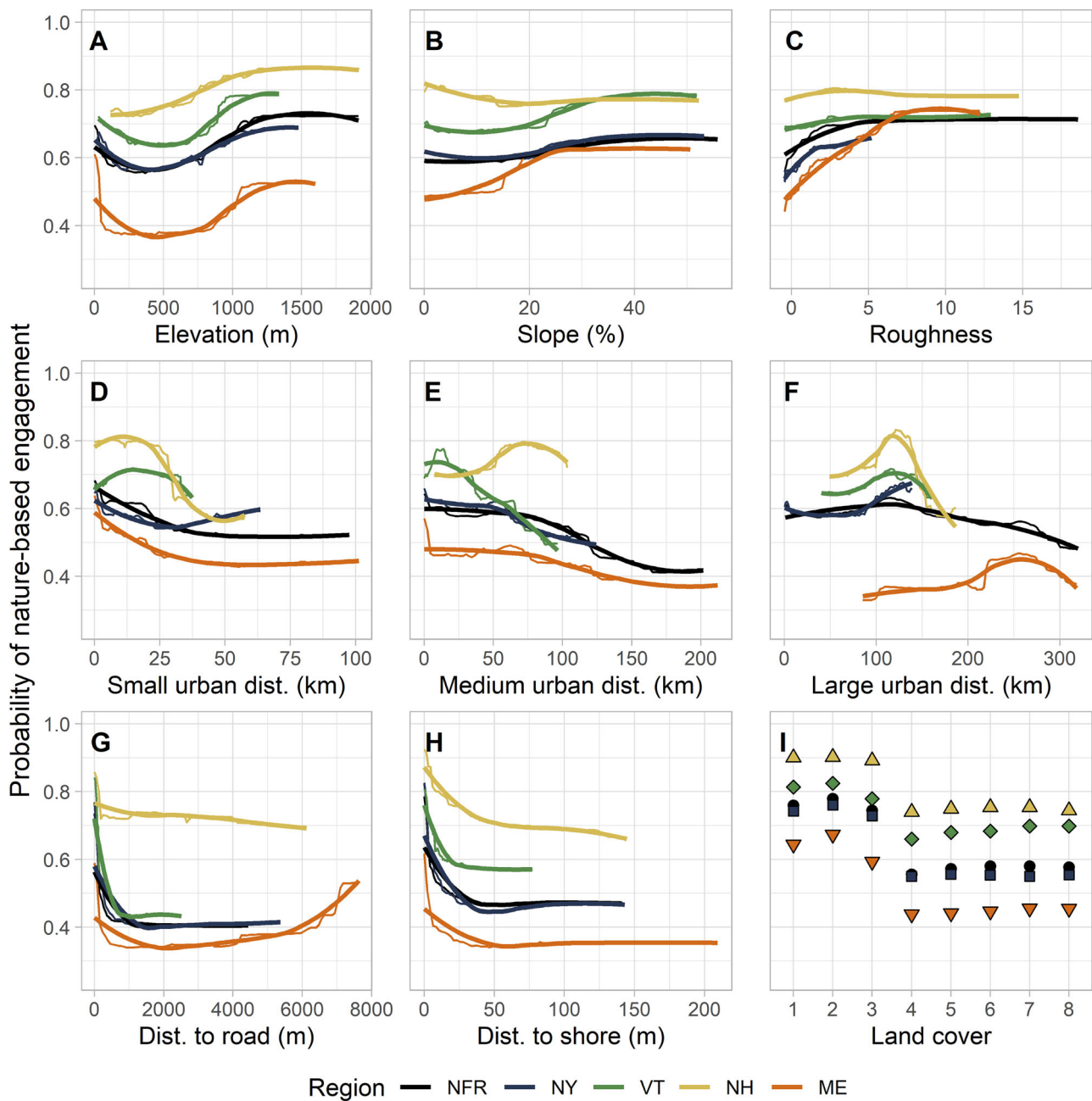


Fig. 5 Relationships between geographic variables and nature-based photography (i.e., CES images) in all rural areas (public and private) of the Northern Forest Region (NFR). Variables include elevation (A), slope (B), roughness (C), distance to small urban areas (D), distance to medium urban areas (E), distance to large urban areas (F), distance to

roads (G), distance to shorelines (H), and land cover (I). Individual partial dependence curves or point estimates are shown for each geopolitical scale overlaid with LOESS regression splines. Land cover classes are referenced in the Methods (“Spatial content analysis”)

areas where engagement with CES features was generally greater at higher elevations, especially in New Hampshire (Fig. 6A). Proximity to urban centers was more important in public areas than the region at large, displaying complex relationships with CES photography in different states, such as New Hampshire, where activity dropped precipitously 50 to 200 km away from large cities (Figs. 5D–F, 6D–F). At the regional scale, CES

engagement gradually declined with increasing distances to small and medium-sized cities, with proximity to small cities ranking as the third most important driver of photography in public areas. Finally, land cover had variable effects on CES engagement, showing pronounced importance in certain states (New Hampshire and New York) where open areas had higher probabilities of CES photography (Figs. 4, 5I, 6I).

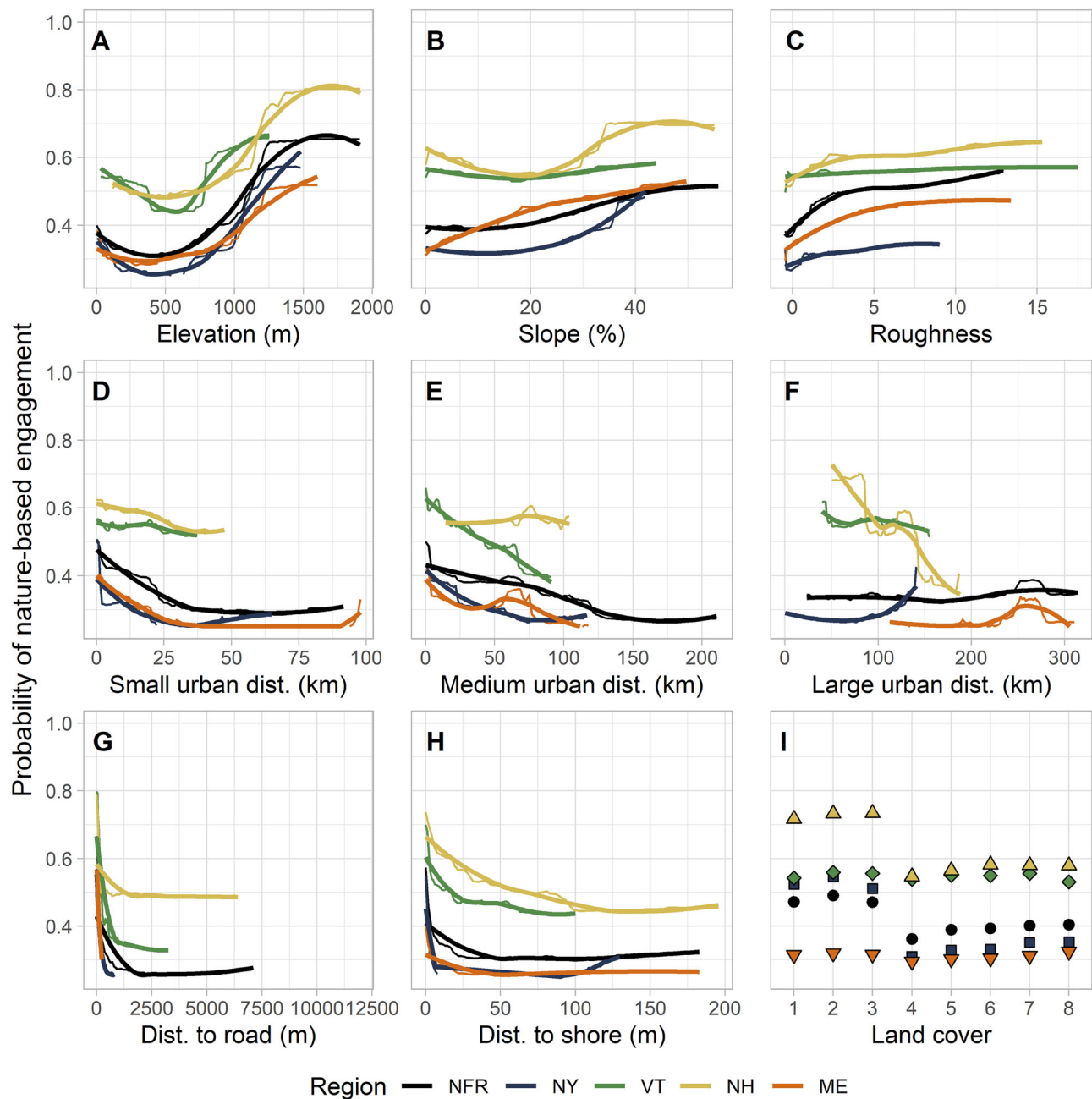


Fig. 6 Relationships between geographic variables and nature-based photography (i.e., CES images) in public areas of the Northern Forest Region (NFR). Variables include elevation (A), slope (B), roughness (C), distance to small urban areas (D), distance to medium urban areas (E), distance to large urban areas (F), distance to roads (G), distance to

shorelines (H), and land cover (I). Individual partial dependence curves are shown for each geopolitical scale overlaid with LOESS regression splines. Land cover classes are referenced in the Methods (“Spatial content analysis”)

Model performance was affected by spatial scale and public accessibility in focal regions (Table 3). Models had high (i.e., >0.825) intraspecific accuracy and sensitivity (i.e., predicting within the same region), but interspecific predictive power was more variable, even between neighboring states (e.g., Vermont to New Hampshire). Broad-scale models (i.e., the entire NFR) were more accurate,

sensitive, and specific than the state-level models. Models trained on all rural images were more sensitive than those trained on rural images in public areas. The opposite was true for model specificity, which was greater in public areas at all scales (Table 3). We used the regional rural and public models to predict CES engagement and suitability throughout the Northern Forest Region at 1-km² resolution.

Table 3 Multi-scale model validation statistics for random forest models of rural images in the Northern Forest Region.

		to				
metric	from	NFR	NY	VT	NH	ME
accuracy						
	NFR	0.891 (0.894)	0.962 (0.968)	0.963 (0.956)	0.964 (0.958)	0.974 (0.975)
	NY	0.876 (0.859)	0.891 (0.901)	0.827 (0.785)	0.784 (0.718)	0.848 (0.815)
	VT	0.811 (0.770)	0.793 (0.798)	0.870 (0.857)	0.755 (0.562)	0.806 (0.809)
	NH	0.822 (0.805)	0.780 (0.725)	0.799 (0.785)	0.877 (0.850)	0.810 (0.826)
	ME	0.815 (0.832)	0.704 (0.795)	0.725 (0.660)	0.743 (0.702)	0.913 (0.923)
sensitivity						
	NFR	0.926 (0.833)	0.975 (0.944)	0.985 (0.954)	0.975 (0.951)	0.979 (0.949)
	NY	0.898 (0.730)	0.926 (0.825)	0.904 (0.748)	0.802 (0.578)	0.871 (0.623)
	VT	0.803 (0.486)	0.791 (0.496)	0.935 (0.869)	0.744 (0.287)	0.778 (0.503)
	NH	0.909 (0.725)	0.904 (0.700)	0.940 (0.768)	0.951 (0.858)	0.848 (0.522)
	ME	0.740 (0.608)	0.573 (0.405)	0.684 (0.419)	0.728 (0.548)	0.906 (0.855)
specificity						
	NFR	0.839 (0.932)	0.942 (0.979)	0.913 (0.959)	0.920 (0.966)	0.971 (0.987)
	NY	0.843 (0.935)	0.834 (0.934)	0.647 (0.830)	0.711 (0.906)	0.827 (0.904)
	VT	0.823 (0.937)	0.796 (0.928)	0.702 (0.842)	0.796 (0.929)	0.830 (0.950)
	NH	0.696 (0.853)	0.588 (0.735)	0.471 (0.806)	0.587 (0.838)	0.777 (0.967)
	ME	0.924 (0.965)	0.907 (0.963)	0.821 (0.949)	0.799 (0.908)	0.918 (0.955)

Performance metrics (accuracy, sensitivity, and specificity) were estimated within and among different geopolitical regions, using testing data from individual states and the entire Northern Forest Region. Public-zone model validation statistics are shown in parentheses

Model predictions reveal patchy landscapes of observed and potential use of cultural ecosystem services (Fig. 7), with hotspots centered in accessible mountainous and coastal regions. Engagement and suitability surfaces were similar at the broad regional scale, with the latter map showing generally higher probabilities, particularly in regions with relatively little visitation (e.g., northern Maine). There were greater contrasts between engagement and suitability surfaces from state-level models, particularly in Vermont and Maine, where large areas of unvisited private lands are moderately suitable for nature-based recreation (Figs. S4–S7).

Validation of Automated Image Tags

The majority of sample images from protected areas (83% and 86% of photographs according to two independent researchers) were related to recreation. There was a 92% agreement between the two researchers in the first validation exercise, suggesting that the recreation content of photographs is unambiguous. Roughly 64% of top tag assignments by Clarifai (63% for the first group and 65% for the second group of students) were found to be related to images in the second validation exercise, with perceptions of tag relevance varying greatly, depending on the image (range of 5–100% agreement by students). Finally, the percent agreement between the researcher-generated codes

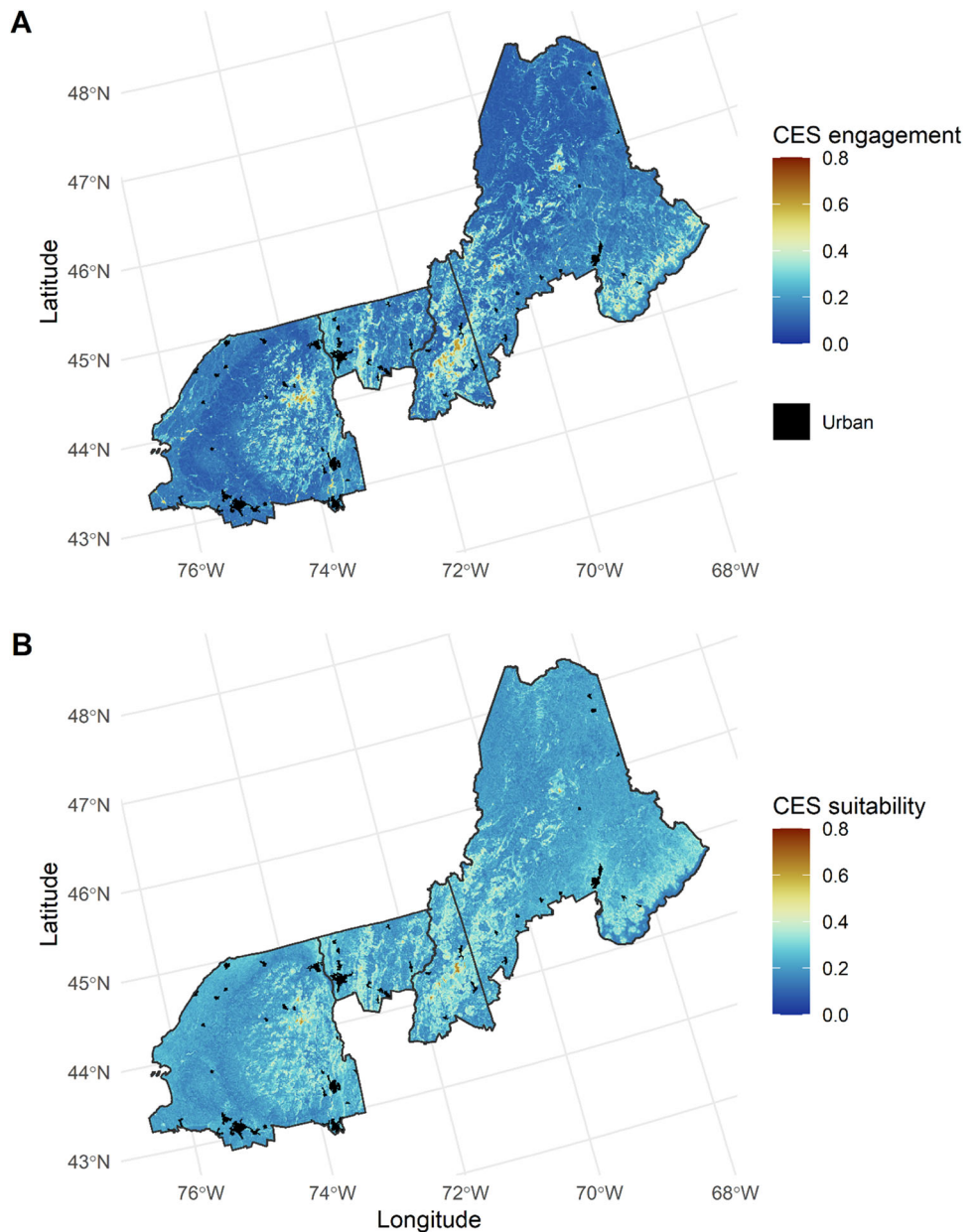
for image captions and the Clarifai-generated tags averaged 87%, with the percent agreement ranging from 66% for “plants” (a generic term) to 99% for “waterfall” (a specific term) (Table S2). These validation results indicate that overall, Clarifai does an adequate job of creating tags that represent the intent of the photographer.

Discussion

Multiple visual themes can be identified in crowdsourced imagery, including several indicative of nature-based tourism and recreation. The most photo- and tag-rich theme in our dataset (scenery) accounted for roughly 60% of all rural images that targeted broad, landscape-level features (panoramas of water, mountains, forests, etc.). A similar result was found by Lee et al. (2019), which estimated roughly 54% of rural images in the Mulde River basin in Germany were focused on “landscape aesthetics.” The predominance of CES-related images in the Northern Forest (69.8%) was close to the amount found by Lee et al. (2019) (65.1%), suggesting that the majority of Flickr images in rural, inhabited regions are capturing human-environment interactions, and thus may function as a general indicator of cultural ecosystem services.

Integrating space, time, and content of crowdsourced photography provided a robust foundation for evaluating

Fig. 7 Predicted engagement (**A**) and suitability (**B**) of cultural ecosystem services (CES) in the Northern Forest Region. Values of pixels (sq km) represent probabilities of nature-based photography from two regional models, one trained on all data from rural areas (**A**) and one trained on data within public lands and waters (**B**)



relevant factors to cultural ecosystem services in the Northern Forest and predicting visitation and conservation potential over a large rural landscape. Generating automated tags and clustering images based on common visual features can streamline a traditionally cumbersome process for describing and analyzing human–environment interactions from crowdsourced data. For example, tens of thousands of images in the Northern Forest were not apparently related to nature or recreation. As is commonplace in large social media datasets such as Flickr, posts of interest are often diluted by non-target content. Manually processing hundreds of thousands of images to delineate which are relevant to researchers and managers is a tedious, labor-intensive process. Automated image classification accelerated this

endeavor and was corroborated by manual interpretation of images, reinforcing it as a reliable method for delineating large volumes of images; however, this method presented some challenges specific to the classification and clustering algorithms that required subjective review and post-hoc adjustments. Our approach builds on previous work using geotagged imagery by considering patterns and processes in visitor engagement at multiple spatial scales. Large rural management regions, such as the Northern Forest, can contain politically and economically distinct areas (e.g., states, provinces) which exert unique pressures on peoples and places, that in turn, affect cultural ecosystem services. Visitors to such regions must pass through multiple filters—social, financial, geographic, and environmental—before

arriving at their destination and engaging in different activities. While some of these filters are difficult to link to crowdsourced data, others are readily available from remote sensing databases, public land clearinghouses, and government census records. By explicitly accounting for these spatial filters in a hierarchical modeling framework, we identified how patterns and processes of human–environment interactions are informed by geopolitical boundaries.

Spatiotemporal Use of Cultural Ecosystem Services

A routine challenge for spatiotemporal problems is deciding what scale to use for an analysis. Often, multiple scales are important to consider for identifying context-dependent relationships and trends over time; however, this strategy can be cumbersome with big data and is not frequently done in assessments of cultural ecosystem services (Heikinheimo et al. 2017; Mancini et al. 2018; Schirpke et al. 2018). While our spatial analysis only focused on recreation in one season over several years, this approach can easily be extended to explore drivers of nature-based activities in other time periods with unique visitation patterns. In continuous rural regions, tourism consortia and other regional visitation and conservation stakeholders may be inclined to either aggregate social media information over large geographic scales or rely on information from neighboring locations to inform management. Though our modeling results mostly support the former approach for the Northern Forest, several inconsistencies in spatiotemporal patterns of use were identified in the region and extrapolating from one state to another should be done with caution. Previous work in the Northern Forest found stronger relationships between photography and visitation at the state level than the entire regional scale (Kuehn et al. 2020), suggesting that CES assessments using crowdsourced imagery may be more reliable for managers when conducted at appropriate local scales.

Drivers of Nature-Based Engagement

Many landscape attributes contribute to spatial patterns in nature-based visitation, and, as we demonstrate, geopolitical context can mediate these relationships. Clearly, accessibility is an important limiting factor of recreation in rural areas, as shown by the high and consistent ranking of road proximity across all spatial scales in the Northern Forest. This is well-documented in protected areas, where photography generally declines farther from roads (Levin et al. 2015; Walden-Schreiner et al. 2018), reflecting a reluctance of most visitors to stray far from infrastructure. Another key access point—trail systems—are important features for visitors (Neuvonen et al. 2010; Walden-Schreiner et al. 2018), but incorporating these fine spatial networks in

models was difficult in our region where such data were not widely available; trails and other access features should be considered in future applications in the Northern Forest and other areas. Access is confounded by aesthetics near shores of open water, another widespread “linear” feature that displayed a regionally strong but less consistent signature in our recreation models, perhaps reflecting tensions between lowland and upland CES features in different states. In large waterbodies, such as Lake Champlain or the Maine coastline, shores may function as both sources of attraction and soft limits on visitation, enticing visitors with natural scenery and recreation opportunities while also serving as boundaries for landbound visitors. Modeling photographs belonging to individual themes, such as “aquatic recreation,” might parse relationships that are more relevant to different managers and help them quantify specific human–environment interactions. Accounting for quality of different environmental features (e.g., water quality or forest health) may be informative for future assessments of CES imagery in rural areas (Keeler et al. 2015). At a larger scale, all of these relationships are conditioned by population centers themselves, which have context-dependent, non-linear effects on visitation, likely reflecting socioeconomic factors that were not explicitly included in this study. These multifaceted limits of accessibility on visitation can be seen as different manifestations of distance decay functions (Hanink and White 1999), where spatial demand for recreation is tempered by fuzzy costs associated with traveling—from homes to parks and from “park” to local features in the landscape.

Themes and Relevance of Images to Nature-Based Recreation

Our validation work found that automated tags adequately represented image content, as observed in previous research (Richards and Tunçer 2018; Lee et al. 2019; Chang et al. 2020). This is the only study we are aware of that considered whether automated tags produced by machine learning algorithms (e.g., Clarifai, Google Cloud Vision) adequately capture the intent of photographers (as inferred from user-generated photo captions) in rural settings. Overall, most forms of visitor engagement are well represented by automated tags and can be reliably translated to broader semantic themes when considering the entire pool of tags associated with an image; however, common visual features (e.g., plants) can result in misalignments between classification (i.e., tag assignment) and engagement (i.e., actual content). Image captions can serve as additional source of validation and semantic detail for complex images with ambiguous tags, but they are not yet integrated into automated classification pipelines that rely solely on visual content. Accounting for multiple semantic dimensions in classification

procedures could enrich CES assessments by capitalizing on the multimodality of social media content (Toivonen et al. 2019); unfortunately, the potential of this multidimensional approach is limited by poor image captioning rates on open-source platforms such as Flickr as well as general inconsistencies in linguistic captioning styles among populations of users. More research on accessing and integrating these additional sources of metadata would be helpful for further clarifying human–environment interactions from crowd-sourced image data.

Conclusion

Social media data have undoubtedly produced valuable information for scientists, policymakers, and natural resource managers interested in sustainable development and environmental conservation (Toivonen et al. 2019; Teles da Mota, Pickering. (2020)). The spatiotemporal patterns of human–environment interactions in the Northern Forest, as in other locations, can serve as important references for land-use planning to both support regional opportunities for recreation and promote conservation by mitigating adverse impacts of visitation (Walden-Schreiner et al. 2018). As with any source of data, geotagged imagery from online platforms has inherent assumptions that are important to consider when translating from patterns to process to practice. Firstly, not all visitors are capturing their experiences with cameras and posting content on the Internet. While robust relationships between photography and visitation have been measured in parks, spatial content analyses assume that online activity is proportional to actual visitors in space (e.g., beyond parking areas). Uploading biases are difficult to ascertain and should be studied with respect to different types of outdoor activities to better understand processes behind presence/absence point patterns and potentially create new opportunities for modeling visitor routes and itineraries over small and large regions. Flickr is one of the most popular sources of crowdsourced imagery data for environmental studies (Ghermandi and Sinclair 2019; Teles da Mota and Pickering 2020), despite other platforms (e.g., Instagram, Facebook) having far more users. Different platforms comprise different communities of users, and these cultural and demographic idiosyncrasies can bias the quantity and quality of environmental content (van Zanten et al. 2016; Tenkanen et al. 2017). Changes in data ownership and storage policies on individual platforms (Flickr was acquired by SmugMug in 2018) will likely impact these types of studies into the future, potentially compromising repeatability and the capacity to monitor spatiotemporal trends. Cross-platform approaches are preferable for capturing different aspect; of user engagement (Samani et al. 2018) but mining multiple streams of images

can be technologically challenging and cost-prohibitive when paywalls limit access (as in our study). New tools for streamlining access to social media platforms (Fox et al. 2020) and classifying images (Winder et al. 2022) may help researchers harness big data and make them more available to managers going forward.

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Compliance with Ethical Standards

Conflict of Interest The authors declare no competing interests.

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